CS 5593 - Independent Project 2:

Sequences & Data Streams

100 points in total 10 questions × 10 points · Due: 11/01 11:59 PM CST

Each question begins with a short passage adapted from a different published research paper. The passage introduces a representation, similarity measure, clustering method, density concept, or anomaly-detection idea.

For each question:

1. **Read** the provided passage.
2. **Code** by completing the **# TODO** sections in `.ipynb` file. All data, imports, helper functions, and scaffolding are provided.
3. **Explain** your results in the **"Your Explanation"** cell using numbers, tables, neighbors, clusters, or plots from your own run.

Rules

1. Modify code **only** at **# TODO** markers.
2. **Do not change** the provided data, helper functions, or random seeds.
3. Explanations must reference your own outputs.
4. Counterfactual questions must be answered relative to your results.
5. You may consult official library documentation.
6. Office hours are intended primarily for installation or execution problems.

Setup

- Python 3.10 or later:
- pip install scanpy leidenalg igraph scikit-learn scipy matplotlib

Grading

Each question is worth **10 points**:

Coding (5 pts each question): Code runs end-to-end, completes the required work, and produces the correct result

Explanation (5 pts each question): Answers the question-specific prompts using evidence from the student's own outputs

Rubric

Coding

- 5 points: Code runs, performs the required computation, and produces the correct result.
- 3 points: Code runs and performs the required type of work, but the final result is incorrect.
- 0 points: Code is missing, incomplete, does not run, or does not produce the required output.

Explanation

- 5 points: Complete reasoning, correct use of course concepts, references own outputs, and answers all comparisons or counterfactuals.
- 3/4 points: Generally correct but missing an important reason, output reference, comparison, or counterfactual.
- 1/2 points: Mostly generic, restates the code, or answers only part of the prompt.
- 0 points: Missing or unrelated.

Coding and explanation are graded separately. A correct result without explanation earns at most 5/10. An explanation without working code earns no coding points.